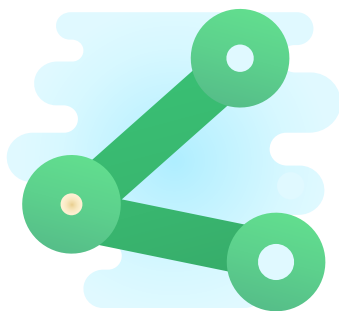


Patuakhali Science and Technology University

Faculty of Computer Science and Engineering

CIT 320 :: Software Development Project-II

Project Report



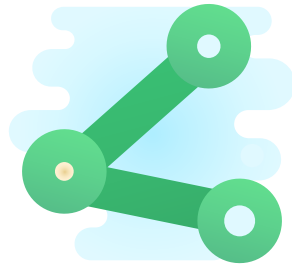
Project Title : Linkerine

Submission Date : 26 June 2026

Submitted from,	Submitted to,
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Linkerine

1. Introduction

Linkerine is a modern, offline first RSS reader and personal knowledge hub that helps users follow, read, organize and sync news feeds, articles, bookmarks, private notes and secrets across all of their devices. In an age of growing information overload, Linkerine combines a progressive web application (PWA) with a local first data layer and cloud synchronization to deliver a fast, installable and fully offline capable experience.

It further integrates optional bring-your-own-key (BYOK) AI for summaries, semantic search and auto-tagging, keyless translation, a zero knowledge encrypted vault for sensitive data and one click following of sources such as YouTube, Reddit, Mastodon and Bluesky.

2. Objectives

1. To build a high performance, responsive application for following and reading RSS feeds, news, articles and managing bookmarks, notes and secrets.
2. To implement a local first offline application with secure cloud synchronization and authentication via Firebase.
3. To provide an “Installable”, fully offline capable experience on mobile and desktop devices.
4. To integrate optional AI for summarization and semantic search, alongside keyless translation.
5. To safeguard sensitive data with a zero knowledge, client side encrypted vault.

3. Problem Statement

Existing solutions for bookmark management are often fragmented. So users must choose between native apps that don’t sync well with the web, or browser extensions that don’t work on mobile. They are often limited to RSS or bookmarks or AI based features with no cross platform support. Some of the solutions also doesn’t support social media sources like YouTube, Reddit, Mastodon or Bluesky.

Furthermore, many modern tools (like Pocket or Raindrop) lock essential features like full text search or unlimited nested folders behind paywalls. There is a need for a unified, open web solution that runs instantly on any browser while offering the performance of a native application with versatile features like offline first operation, cloud sync, AI assistance and secure data storage.

4. Literature Review

According to Z. Chang's et al's [1] survey of modern crawler methods, web crawlers are essential for automating data retrieval from the vast and unstructured internet, which can be classified into various types based on their strategies and targets, such as generic, focused, incremental and distributed crawlers. The choice of libraries, databases and frameworks for building these systems depends heavily on the specific target data, purpose and the programmer's ease.

Later on in "The News Crawler: A Big Data Approach to Local Information Ecosystems" by Khanom et al. [2] they address the growing crisis of losing access to local journalism. Because manually auditing local news coverage is expensive and time consuming. So the authors propose a scalable, semi automated pipeline with a custom web crawler alongside Machine Learning and Natural Language Processing, such as the spaCy library, to evaluate the content.

But due to lack of time and resources, the crawler was unable to capture all content from large, dynamically updating news sites. So in "Using RSS to Improve Web Harvest Results for News Web Sites" by Jones & Neubert [3], strategy has shifted to crawl the sites' RSS feeds twice daily to build targeted, seed lists of newly published URLs but created a navigation problem, as the pages often lacked hyperlinks.

To further improve the user experience and information overload, "A Study on Recommendation Features for an RSS Reader" by Ji & Zhou [4] proposes integrating recommendation algorithms directly into RSS readers. This study worked on: Text Similarity, Favorite Fraction, Inverse Update Frequency and PostRank. The most effective approach was a hybrid feature combination, combining Text Similarity with the Favorite Fraction.

Finally we arrive at "AI driven web crawling for semantic extraction of news content from newspapers" by Saravanan & Ahamed [4]. Traditional crawlers rely on rigid DOM tree rules and keyword matching, which may fail with complex layouts. This paper introduces WISE, an advanced framework that integrates Deep Learning (CNNs and RNNs) and NLP (like BERT embeddings) to understand the semantic context. But it is resource intensive and requires significant computational power and so, less feasible for smaller scale applications.

5. Related Commercial Projects

1. Mozilla Pocket [5] was a popular read it later app, but has become bloated with premium tiers. It was not necessarily a RSS feed reader, but it allowed offline reading and synchronization with mobile devices. As of January 2026, it is no longer actively maintained and has been deprecated.
2. Goodlinks [6] has an excellent UI, but exclusive to the Apple ecosystem. It's not ported to Android or Windows and lacks a web interface. It's relatively expensive than others as well. And it also lacks vault support.
3. Futurwise [7] is a Chromium only extension; lacks a standalone mobile interface. It is not open source and has no offline first capabilities. It is also a paid service. It also lack Firefox/Safari support and non FOSS.

4. Instapaper [8] is Great for reading, but poor for organizing bookmarks. Became the official replacement for Kobo e readers. It doesn't come with vault or AI features and is not FOSS.
5. Raindrop.io [9] is the industry standard, but lacks self hosting capabilities and anonymous usage. It is also limited to bookmarks and does not support RSS feeds. It is also a paid service and not FOSS.

6. Scope

Linkerine is primarily developed as a modern web application using the Next.js framework. The application focuses on server side RSS/Atom parsing, offline first storage, cloud sync and authentication support. It currently natively support news, articles and blogs from RSS supported blogs, hugo, astro, wordpress, SSG and other dynamic platforms where RSS/Atom feeds are available.

For mobile devices, PWA (Progressive Web App) is attached alongside a responsive interface was built with Tailwind CSS and DaisyUI.

The project has since been extended with optional BYOK AI features, a zero knowledge encrypted vault, one click following of social sources (YouTube, Reddit, Mastodon and Bluesky natively) and companion browser extensions that share the same deployment.

It can also be optionally integrated with RSSHub instances for broader social and platform coverage as well.

6.1. Job Market Analysis

The job market for full stack web developers in Bangladesh is growing rapidly, with significant demand for Next.js and modern JavaScript expertise. According to major job portals:

Job Portal	Link	Stack
BD Jobs	1. https://bdjobs.com/jobs/details/1449039?ln=1 2. https://bdjobs.com/jobs/details/1449038?ln=1 3. https://bdjobs.com/jobs/details/1452423?ln=1	React, NodeJS, mongoDB, mongoose, JavaScript, Github, Git
BD Tech Jobs [10] & Techn-Talents [11]	1. https://authlab.io/wpjb-jobs/senior-software-engineer/ 2. https://riseuplabs.com/jobs/full-stack-developer-python-django/	Vue.js, React
eJobs Bangladesh [12] & JobMatchingBD [13]	1. https://www.ejobs.com.bd/jobs/san-francisco-ca-senior-full-stack-developer 2. https://jobmatchingbd.com/job/engineering-jobs-14/ 3. https://jobmatchingbd.com/job/it-jobs-in-bangladesh-second-source/	Node.js, React/ Vue, PostgreSQL or, MongoDB

Arc.dev [14] & Wellfound [15]	1. https://wellfound.com/jobs/2993781-senior-react-and-nextjs-web-developer 2. https://wellfound.com/jobs/2927551-senior-full-stack-developer	React, npm, TypeScript, Next.js and Git
Careerjet Bangladesh [16]	1. https://www.careerjet.com.bd/jobad/bdbe02ffc4686d4de58c19bc2c53c6f999 2. https://www.careerjet.com.bd/jobad/bda7f026ae3ccb8bd6ab7d9027f8d5c26f	React, NodeJS, mongoDB, mongoose, JavaScript, Github, Git, Tailwind CSS

Table 1: Job Market Opportunities for Next.js & Full-stack Developers in Bangladesh

The demand for full stack developers with Next.js expertise is becoming the standard for modern web development in Bangladesh, making Linkerine a timely project that demonstrates bleeding edge skill technology.

7. Methodology

7.1. Technology Stack

The development of Linkerine will follow an agile methodology. We have selected a stack optimized for performance, type safety and rapid deployment.

Component	Technology
Frontend Framework	Next.js 16 (React 19, App Router, Turbopack)
Language	TypeScript
UI/ UX Design and Icons	Inkscape vector graphics (SVG)
Styling	Tailwind CSS 4 + DaisyUI 5 (with Radix UI primitives)
Backend / API	Next.js Route Handlers (SSRF-protected feed proxy)
Cloud Database	Firebase Firestore (user data, multi-device sync)
Local Storage	Dexie (IndexedDB) — offline article cache
Authentication	Firebase Authentication
AI (optional, BYOK)	OpenAI / Google Gemini / Anthropic Claude
Encryption	Web Crypto API (AES-256) for the vault
Hosting	Vercel (frontend), Netlify (Backend), GitHub (VSC)

Table 2: Technology Stack for Linkerine

7.2. Design Principles

The design of Linkerine will adhere to the following principles:

1. Ensuring the layout adapts perfectly from mobile screens to large desktop monitors using Tailwind CSS and DaisyUI, with installable PWA support.
2. A Dexie/IndexedDB cache keeps articles and the interface fully usable without a network; Firebase synchronizes user data across devices when online.
3. AI keys are stored only in the browser (BYOK) and never sent to the app's servers, while the vault is end-to-end encrypted (zero-knowledge).
4. Using TypeScript to ensure robust data validation from the database to the UI.

8. Feature Comparison

Table 3 contrasts Linkerine with widely used and commercial read-later, bookmarking and RSS tools.

App	Dev	RSS	Of- fline	AI	Vault	FOSS	Web	Desk- top	Phone	Chrome	Fire- fox
Link- erine	✓	✓	✓	✓	✓	✓	✓	✓	not iOS	✓	✓
Mozilla Pocket	Dead	—	✓	—	—	✓	✓	✓	✓	✓	✓
Rain- drop	✓	✓	Paid	—	—	—	✓	✓	✓	✓	✓
In- stapa- per	✓	—	✓	—	—	—	✓	Mac only	✓	✓	✓
Good Links	✓	—	✓	—	—	—	—	Mac only	iOS only	—	—
Feedly	✓	✓	Paid	Paid	—	—	✓	—	✓	✓	✓
Futur- wise	✓	✓	—	—	—	—	—	—	✓	✓	—

Table 3: Feature comparison of Linkerine with existing solutions across platforms

9. Visual Models

9.1. ERD (Entity Relationship Diagram)

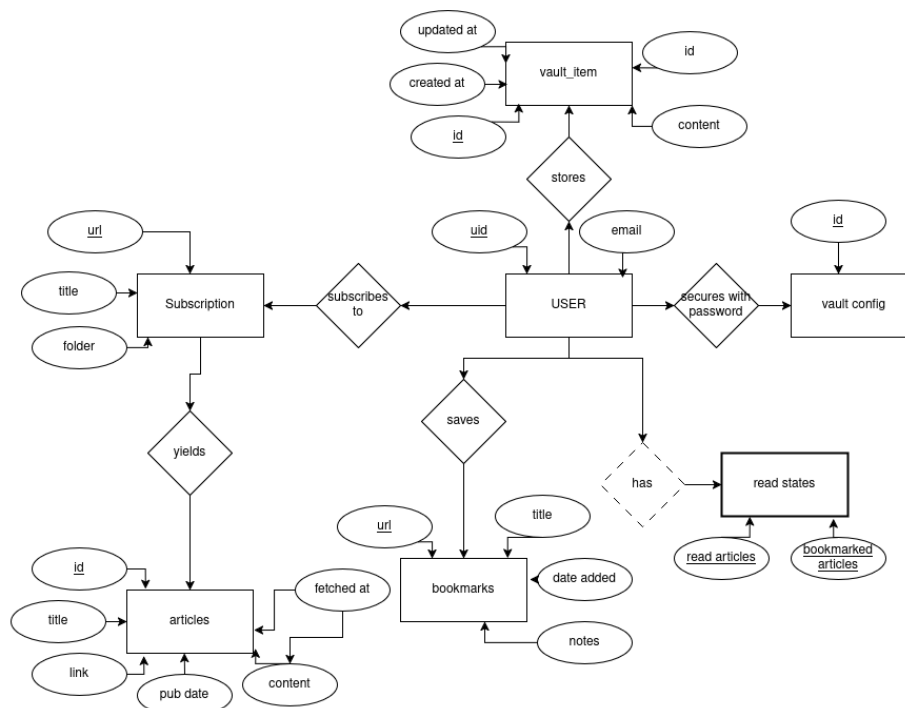


Figure 1: Entity Relationship Diagram of Linkerine (Firestore + Dexie)

9.2. Schema Diagram

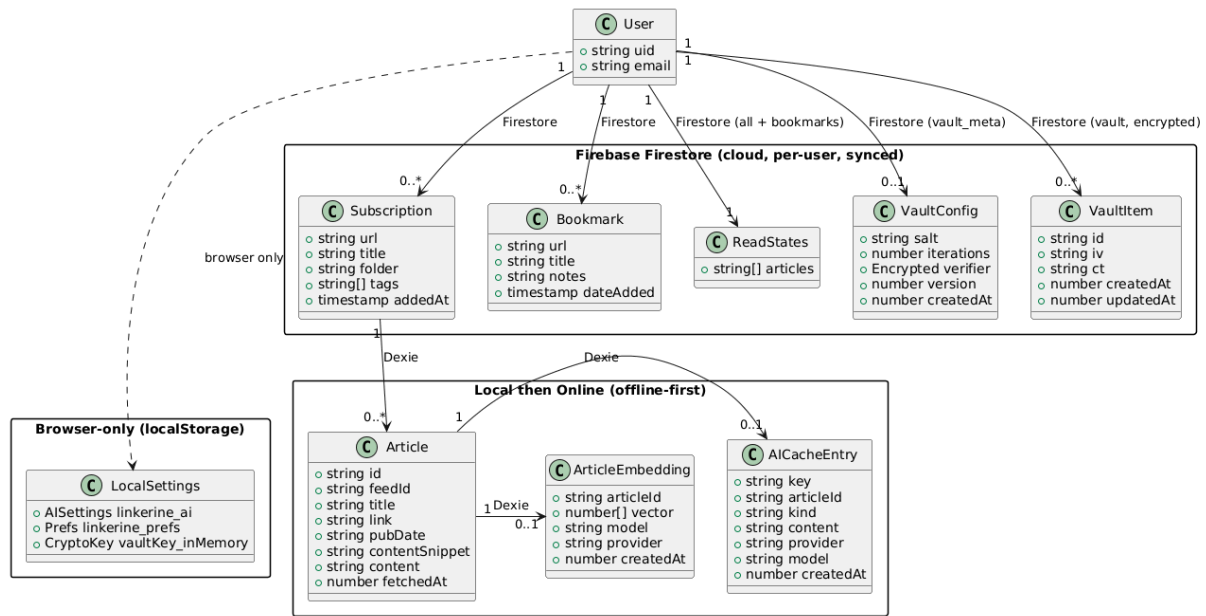


Figure 2: Storage schema — Firestore (cloud sync) and Dexie (offline) records

Figure 2 shows the overall schema diagram of Linkerine.

9.3. Data Flow Diagram

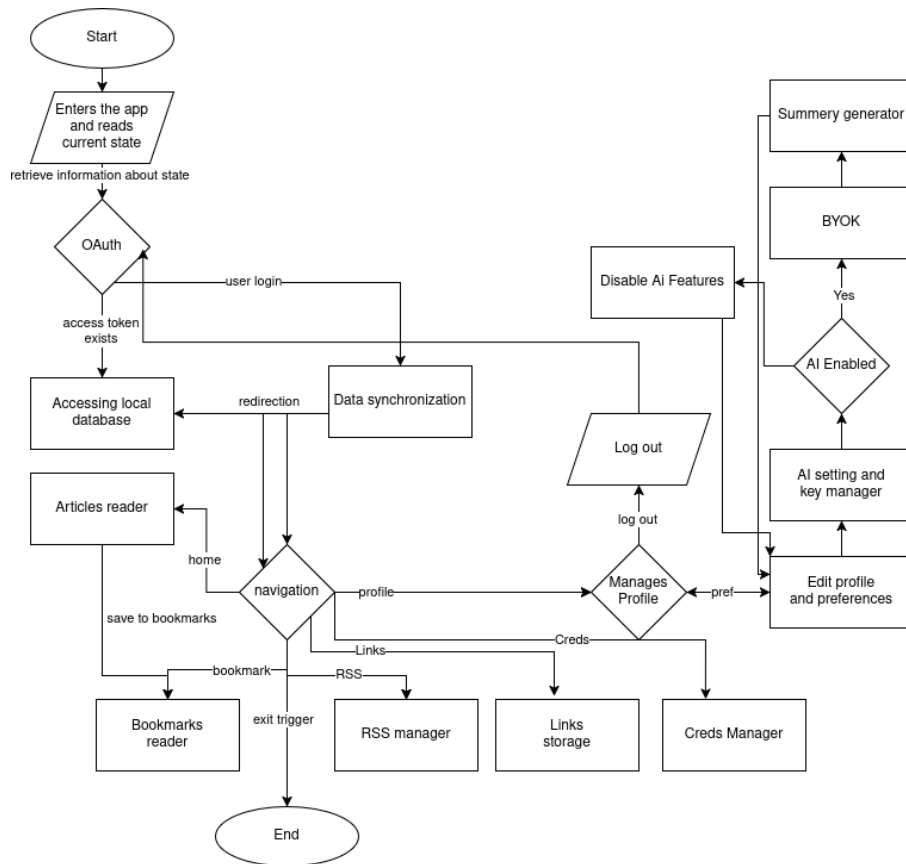


Figure 3: Source-addition and reading data flow in Linkerine

Here Figure 3 illustrates the architecture and flow chart of Linkerine.

9.4. Timeline (Gantt Chart)

The timeline is divided into 12 weeks, focusing on the Next.js development lifecycle.

Task	Week 1-2	Week 3-4	Week 5-6	Week 7-8	Week 9	Week 10	Week 11	Week 12
Setup Next.js 16, TS & Firebase Auth	✓		✓					
Data layer (Firestore + Dexie/IndexedDB)		✓	✓					
Feed sync proxy & subscriptions			✓	✓				
RSS/Atom parser & reader views				✓				
Search, folders/tags & BYOK AI			✓	✓	✓			
Encrypted vault & social sources						✓		
PWA, DaisyUI themes & UI polish						✓	✓	
Vercel deployment & testing								✓

Table 4: Development Timeline of Linkerine

10. UI Mockups

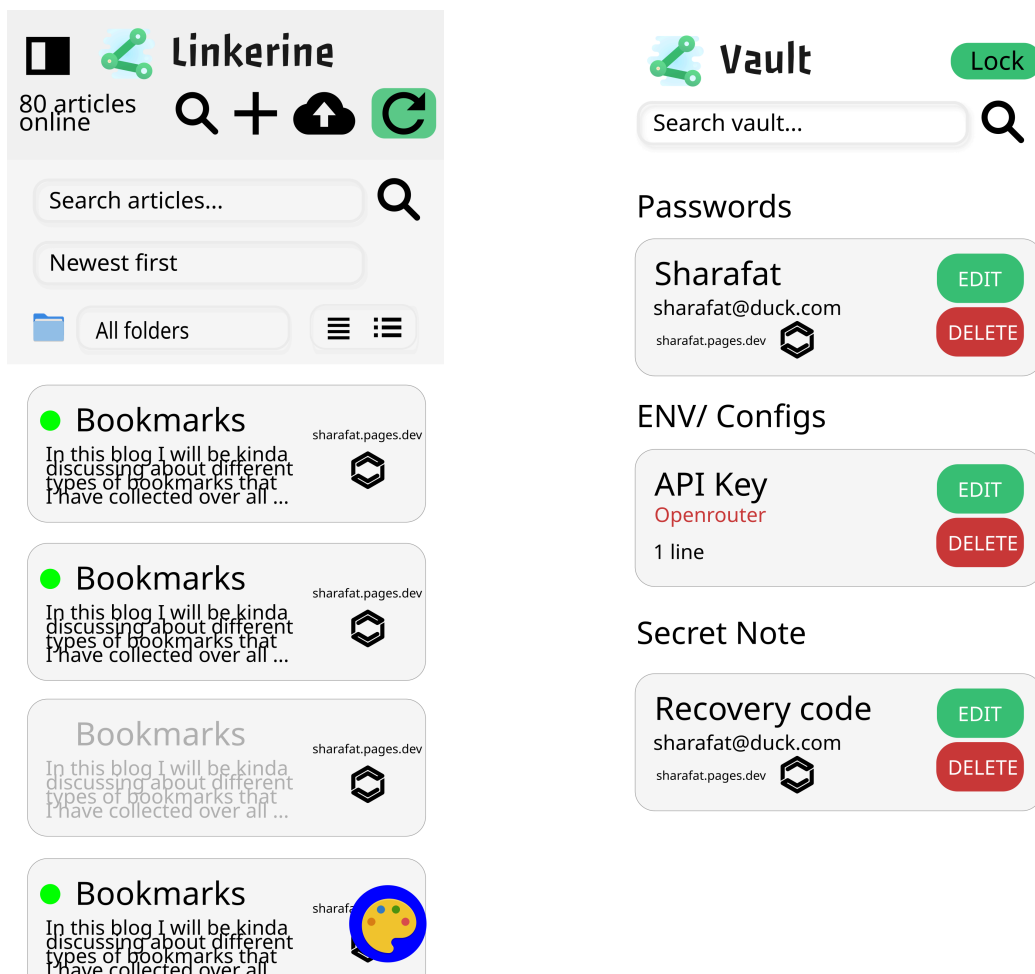


Figure 4: Landing Page and Vault

Bookmarks are the general way to store links and other info.

Jan 25, 2026

My personal bookmarks collection. Bookmarks of this list can be outdated as well as corrupted. Feel free to fork this project and create your own!

If you can't forget the name of a website then that site doesn't deserve to be your bookmark. And with that quote of mine I avoided social sites and other sites that you can't forget easily! For example, Facebook. Now my question is why can't we forget it? Let us know in the comment. And, finally, special thanks to all of

AI tools Hide

Translate

Chinese

Summerize

Translate

Ask about...

ask

sharafat.pages.dev 

Discover Feeds



Bangladesh

● Bookmarks

In this blog I will be kinda

subscribe

sharafat.pages.dev



Bookmarks

In this blog I will be kinda

done

sharafat.pages.dev



● Bookmarks

In this blog I will be kinda

subscribe

sharafat.pages.dev



Linux

● Bookmarks

In this blog I will be kinda

subscribe

sharafat.pages.dev



Figure 5: Reading view and Discover Feeds view

Your Feed
86 articles offline

Q Explore + Add Feed Import Read All Digest Sync

Newest first All All feeds All folders

PM leaves Dalian for Beijing by bullet train

After attending the World Economic Forum (WEF), Prime Minister Tarique Rahman today, Wednesday...

en.prothomalo.com Jun 24

Animated wallpaper and widgets

I installed Gnome Live Wallpaper but I realized that now my Desktop Widgets are hidden when the...

www.reidill.com Jun 24

Housing sector: New tax burden raises fears of another spike in apartment prices

Dishi Properties is currently developing eight housing projects across different parts of Dhaka, six of...

en.prothomalo.com Jun 24

Iran's top diplomat speaks with Hamas

Iran's top diplomat spoke with a senior official from the Palestinian militant group Hamas regarding tal...

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Shajgoj: Building trust before selling products

Long before beauty tutorials filled social media feeds in Bangladesh, finding reliable skincare advice...

en.prothomalo.com Jun 24

Discomfort over unhygienic toilets at school

Lions Model School, located in the Nayatolia area of Moghbazar in the capital, provides education up t...

en.prothomalo.com Jun 24

Navana Group vice chairman Sajedul Islam arrested

Police have arrested Sajedul Islam, vice chairman of Navana Group. Officers arrested him from his...

en.prothomalo.com Jun 24

BB introduces new account facility for expatriate Bangladeshis

Bangladesh Bank has introduced a new banking facility for expatriate Bangladeshis. The facility is...

en.prothomalo.com Jun 24

PM Tarique Rahman takes WEF summit to Police sta...

en.prothomalo.com Jun 24, 2026

Iran's top diplomat speaks with Hamas

Iran's top diplomat spoke with a senior official from the Palestinian militant group Hamas regarding talks between Iran and the United States, state television reported on Wednesday.

Abbas Araghchi discussed the "latest developments" in the region with Basem Naeem, a member of Hamas's political bureau, Iranian television said.

The call follows the signing last week of a memorandum of understanding between Iran and the United States aimed at bringing a lasting end to hostilities in the Middle East.

The text does not mention Gaza but stipulates "an immediate and permanent cessation of military operations on all fronts, including in Lebanon".

Hamas has welcomed the agreement and expressed hope that it would also help end the violence in the Gaza Strip, which has been devastated by over two years of war with Israel.

Iran is a staunch supporter of the Palestinian cause, making it a cornerstone of its foreign policy since the 1979 Islamic Revolution.

During the call, Araghchi "reaffirmed the Islamic republic's continued support for Palestinians and their just cause until their legitimate national rights are fully realised", according to a transcript reported by Iranian television.

On Tuesday, the Iranian foreign minister accompanied President Masoud Pezeshkian on a visit to neighbourin Pakistan, which has been acting as a mediator in the talks between Tehran and Washington.

Figure 6: Landing page in Desktop view's screenshot

10

11. Security and Privacy

Our app utilizes offline first, client-driven architecture. It secures user data through Firebase Authentication, strict Firestore access rules and client-side zero-knowledge encryption for sensitive information.

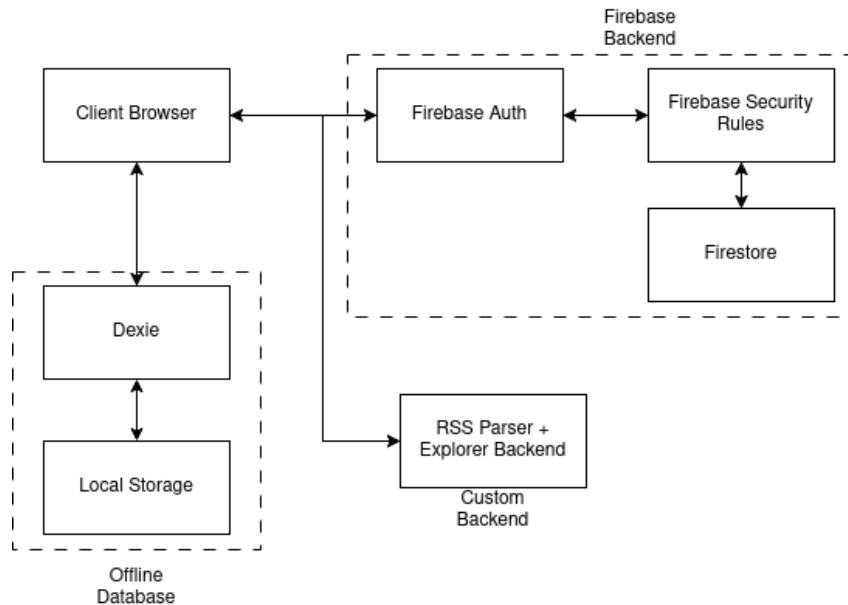


Figure 7: Security and Privacy model of Linkerine

Here in Figure 7, we store all sensitive data in a zero-knowledge encrypted vault. Furthermore fetched articles are stored only in the browser's local storage and are never sent to the server. The Firebase Security Rules are used to protect from unauthorized access and SSRF attacks.

12. Future Plans

Several capabilities have been implemented, but the following remain as planned extensions:

1. To allow users to publish or share specific feeds or folders as public, shareable read-only web pages.
2. To enable native push notifications and periodic background synchronization to surface new articles even when the app is closed.
3. To allow more broader social and platform coverage, with first-class support for self-hosted RSSHub instances.

13. Result

The outcomes of the project are as follows:

1. The project has successfully implemented a modern, offline-first RSS reader and personal knowledge hub with a local-first data layer, cloud synchronization, optional BYOK AI features and a zero-knowledge encrypted vault.
2. Semantic search and keyless translation have been integrated, enhancing the user experience and accessibility of content.
3. The application has been deployed on Vercel and netlify, ensuring high performance and availability across all device types.

4. The project has been designed with a focus on privacy, security and user control, allowing users to manage their digital content without compromising their personal data.

14. Conclusion

The project has successfully demonstrated the development of a modern, offline-first RSS reader and personal knowledge hub that addresses the limitations of existing solutions. It's browser extension will help users to more easily access our website. The vault and BYOK AI features provide additional value, enhancing the UX and ensuring privacy and security. Bookmarks and links can be organized and managed effectively, while the optional AI features offer summaries and semantic search.

As of now, most of the blog sites, news sites and some social media platforms support RSS/Atom feeds, making Linkerine a versatile tool for managing digital content. The project has been designed with scalability in mind, allowing for future enhancements and integrations. Currently PWA is supported on Android, Windows, Linux and MacOS. iOS support is limited due to Apple's restrictions on PWA features.

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